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Themes in Metaphysics

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On the Role and Significance of *Dunamis*/*Energeia Kata Kinesis* in Aristotle's Sublunar Natural Ontology

The philosophy of Aristotle emerges from a milieu still working in the fallout of the original Parmenidean insight. The experience of many *beings* suggests the existence of a unitary *Being*, but the reconciliation of this insight with the variegated phenomena of the world proves no small task. The phenomena require an altogether project of categorization and prioritization in order to be resolved, but the determinations involved are exceedingly difficult to secure, even while tending to cleave along certain basic arrangements. Aristotle signals a new strategy for development. The being of any given being is not spoken of simply, but with respect to a distinct opposite. Moreover, there are four ways of opposition that beings seem to admit: whether *being a present thing* (or attached to a present thing) or *being an absent non-thing*; whether *being-at-work* or *in potency*; whether *being true* or *false*; and whether *being incidental* or *pertinent*. These ways indicate some kind of interrelation and unifying meaning, but not as species common under a genus, which would again eclipse their evident difference, but as members sharing in an ambiguous analogical resemblance. The products of this development are manifest in Aristotle's solution to the problem of motion, a phenomenon he charges virtually all of his predecessors with failing to have grasped. Motion is the being-at-work of a potentiality *qua* potentiality. The understanding of motion, then, is contingent on the understanding of *dunamis/energeia,* their way of being, and the relation of that way to the others. As a way of approaching and illustrating this cluster of phenomena, I propose to establish Aristotle's ontology of sublunar nature (nature, "the origin of all those beings which bear in themselves an origin of *motion*").

The approach of this essay, attempting to draw into relief Aristotle's analysis at *Metaphysics Θ1-3* and Heidegger's interpretation thereof, is (1) to build-up Aristotle's ontology of nature, including human nature, and to clarify the roles of *dunamis, energeia,* and *kinesis*; and (2) to outline, in terms of this clarification, the opposition posed by Aristotle's contemporaries, the Megarians and the Protagoreans.

1.

In the first book of the *Physics*, Aristotle establishes the basic structural principles of nature. We notice that all change occurs along vectors of opposition: we indicate something as white in virtue of its lack of blackness, and not in virtue of its lack of education (188a35). White ceases to be to the extent that it becomes black, and black comes into being to the extent that it ceases to be white, through a continuum of intermediate conditions. This principle of change applies to all things: as white is in opposition to black, for example, the concordant house is in opposition to the disconcordant house, i.e., the housing materials (188b15). This principle alone, however, does not yet articulate the manifold dependencies we witness in nature.

By cross-referencing the various ways we speak of something changing into another, two additional principles become apparent. We differentiate, among complex things, such as “an uneducated person”, those aspects which persist through a change (“person”) and those which are destroyed through a change (“uneducated”). Within a complex, one aspect can be dependent on another. The first principle this suggests is that we are somehow able to pick out certain phenomena as independent *things* (191a13-14). For instance, if an uneducated person becomes an educated person, we register the generation of education as dependent on the underlying thing, the person; whereas if a white house was said to have somehow become a white statue, we would not make the fence and the sculpture dependent on the whiteness. We have an innate ability to see things in definition and to distinguish them from non-things, e.g., predicates. The second principle is that one aspect can simultaneously underlie another. Not only are various predicates underlain by the thing they predicate, but reflection shows that this thing too is dependent on the existence of an other something (190b3-6). This principle allows for dependency between things, e.g., between a finished house and its bricks and frames. The first principle, that change follows vectors of opposition, still applies, but with new body: the movement of predicates along their oppositional vectors are "horizontal" or reciprocal in character, coming-into-being or out-of-being *qua* their own opposition, but not strictly speaking affecting the being or non-being opposition of their underlying *thing*. The being or non-being opposition of the underlying *thing* follows a "vertical" vector of stacked dependencies. As it ceases to be itself, it comes to be its non-being *qua* itself; that is, it returns to its underlying material, its own peculiar non-being (188b15).

At base of the dependencies of things, must exist a finite number of perceptible, namely tangible, things (*Gen. et Corr.* , 329b6-10). Those are in error who postulate the base as only one thing: what mechanism would this leave for the proliferation of things (329a7-11)? It may be necessary to postulate some primary material to facilitate the change from one into another, but in any case such material would be inseparable from the elemental things (329a30-31). Aristotle arrives at four simple elemental things (such-as-fire, such-as-earth, such-as-air, such-as-water) (320b24-26). They are things, but naturally exist along reciprocal vectors (hot/cold, wet/dry) so that one's coming-into-being is the other's very ceasing-to-be, fueling perpetual generation and obliteration (331a13-17). As the elemental things gradually transform into their opposite, they more or less cross-over the other vector of opposition, resulting in four possible couplings. At this elementary level, things are such that their "vertical" or dependent vector of opposition – between their being-themselves or reverting to their underlying thing – all but collapses into a "horizontal" vector of reciprocal transformation occurring upon a surface.

Aristotle proposes the following solution for how compounds like flesh and bones are formed from this elemental churn. Whenever any of the elements is in the process of transforming it exists as a mean combining the opposites at a certain ratio (334b9-40). While the elements are in the process of shifting from one thing to its opposite, this exchange fosters the generation of the product of their combined material state, insofar as the appropriate mean condition holds. This coming-into-being of the mean state of contradictory things is appropriately called coming-into-being and not a complex alteration of prime matter in virtue of the independent "thingly" character of the objects generated. We register these combinations not as mere mixtures of elements, but determinate objects unto themselves. Somehow, nature bestows such thinghood on elemental bodies coordinated in “such-and-such a determinate condition” (333b17-18). These can at once be presumed to depend somehow on their underlying elemental arrangement – if such-as-water completes its transition to such-as-fire the thing constituted out of their mean would cease to be – but also to somehow be more than the sum of their parts, to constitute a “uniform” and “regular” structure (333b5-8), which can be spoken of independently, and even with regards to supervening dynamics. For instance, the bone would not only be destroyed by the completion of the transition from water to fire, but also by the impact of a heavy rock. The elemental things then, in their perpetual reciprocal generation and destruction, act as a substrate for the emergence of compound things.

The compound things literally embody collections of elemental things in processes of transformation along their oppositional vectors, insofar as they remain with certain mean boundaries peculiar to the constitution of the given thing. Looking at the elemental things simply, we would understand things to exist in perpetual transformation into opposites. Looking at the compound things simply we see that, in one sense, they are subject to or dependent on such elemental transformations, but, in another, their embodiment allows for a new degree of relational differentiation. “[O]ne independent thing is not contrary to another independent thing” (*Phys.*, 189a35). Compound things maintain non-reciprocal relationships among each other, relations indicative of a variety of possible transformations. A rock can knock other rocks into different places without destroying them.

It is characteristic of the compound things that they can be shaped, supplemented, sculpted, composed, or altered in order to bring other compound things into being (*Phys.*, 190b5). For example, water and earth in a certain proportion produce clay. The clay-thing can be formed into a brick-thing. And we could, moreover, combine many brick-things with frame-things and so on in order to bring a house-thing into being. In other words, compound things are highly order-able and "stackable" into other things. This kind of generation of things, however, has an ersatz quality, insofar as such combined compounds are especially susceptible to the suspicion that their thinghood does not reflect a determinative configuration written into the fabric of nature but one dependent for its unity on the human subject. A house would be no more than the sum of its parts; an arrangement of its basic things into a heap (*Met.,* 1041b28-30). The significance of this aspect of compound things must be left until the discussion of those kinds of things capable of producing them.

Things which *live* are more resistant to such reductive suspicions. We pick out things as living when they are able to nourish themselves and grow and decay, or when they show any of the following: intellect, perception, free movement (*DA*, 412a13-15). Such things are more than compound things; while they may depend on a compound thing in such-and-such a determinate condition, the linking of the highly distinctive phenomena of life with a compound thing suggests that they constitute a new *kind* of thinghood (412a16-20).

The thinghoods of living things ascribe to a tripartite hierarchy, with each higher stratum building on capacities established at the level below. The first level is proper to plant-life but present throughout as the nutritive potentiality, that in virtue of which life is at all. Characteristic of this level of thinghood is the possession of the reproductive and the nutritive capacities (415a25). The nutritive refers to the ingestion and transformation of surrounding matter as a way of sustaining itself. Compound things are sustained inasmuch as their elemental substrate holds within a given determinate mean. The nutritive capacity, then, involves an active engagement with the plant-thing's surrounding things (namely water-and-earth rich soil), such that it is able to co-opt the preservation of their thinghood, their elemental configuration, for the preservation of its own, responding to given vectors of configurational vulnerability (e.g., increasing Fire). It is not only able to intake its surrounding things, but to hasten and alter their decomposition as necessary. The reproductive refers to the soul's initiative to sustain itself in perpetuity, by producing seeds bearing the potential to achieve the same determinate configuration as the plant-thing, but in an other thing. This occurs, presumably, by way of a self-modification program, along with the accumulation of material by continued intake of nutrients. In this most basic form of vital thinghood we see a new degree of involvement with the surrounding realm of things. Plant-things are related to their proximity such that they themselves can reach out and transform their surroundings (following two highly effective strategies) in order to further the preservation of their thinghood.

The second tier is marked by the arrival of things with sense perception, namely, at the minimum, tactile perception, this being “a sort of mean between the opposites present in objects of perception” (424a4-5). This is a localized calibration of a frequency of opposition which is drawn to one or other extreme based on the qualities of proximal things. As animal-species escalate in sophistication, they gain up to five such calibrations, each of which layers a different dimension onto the original tactile. (This is to say that taste, scent, hearing, and sight are not altogether different from the perception of the tangible, but each attune a mean according to tangible things *qua* a peculiar vector of appearance, and the latter three “through” relatively intangible elemental mediums.) The respective calibrations are reconciled by a common sense which, in its drawing to unification, is able, with a measure of error, to assess whether varied phenomena belong to a single thing, and to act as a second-order mean for deriving common phenomena (movement, rest, number, figure, size), incidentally transmitted through the respective senses (418a16-18). The senses, then, have the effect of differentiating and sharpening the animal-thing's surrounding things, beyond the plant-thing's comparatively blind groping.

These localized calibrations produce a cluster of interrelated developments. Pleasure is the state of the calibrations such that they signal the animal's apparent good (431a10-12) and desire is the absence of having achieved this (414b4-5). The animal's movement is governed by the object of desire – it is always in pursuit – and the object's presence is conveyed to the animal by the imagination (433b7-9). The imagination grows out of the unreliability of certain classes of sense-impression, which thereby produce calibrations misconstruing the surrounding phenomena the animal is constantly subject to, building a disparity between the surrounding world and the sensory data the animal is compiling (428b17ff.). This gap between perceiver and perceived is harnessed as follows. When sense perceptions affect the senses, the perturbations do not stop immediately (424a17-20). To the extent that an animal is able to grip and retain such impressions (and this is memory), it becomes conditioned to similar arrangements of surrounding things and associates them with a history of thwarted or consummated desires (cf. *APo* 100aff.). It thereby builds a bifurcation between the apparent good characteristic of the innate tendencies of its species and the apparent good given the adjusted calibrations. This latter may align with the former, as when an animal comes to learn the pathways of its surrounding locale, or may exist in tension, objects of desire being charged with new ambivalence, as in the case of a dog trained not to climb onto a bed but baited by a treat (434a12-14). The animal builds a microcosm of its phenomenal surroundings, carved with affective charges suggesting efficient and inefficient ways to acquiring its objects of desire. In pursuing their prey, animals open to a pressure to submit their immediate perception of surrounding phenomena to a generalized perception gathered over time.

The uppermost tier is characterized by the mature capacity for *logos* and understanding. Out of the continued development of the phantasm-layering capacity, comes not only experience -- the tendency toward universalizing common features – but knowledge (*APo*, 100a5-7). Knowledge is, somehow, to jump to the terminus of a definition; the determination of a thing which holds in all cases (*Met.,* 981a4-14). The understanding depends on phantasms as though they were sense-impressions, and constitutes not a passively received affection but a sort of preservative state change (*DA,* 431a13-14), akin to disorganized soldiers setting into formation (*APo,* 100a10-14). Thinking occurs by the free play of retained phantasms. The human uses words as a way of symbolizing previous experiences into discrete phantasms (*DI,* 16a4-9) and dancing them across the affects as a way of simulating states of affairs from various speculative perspectives. These complex unities must then be verified for their correspondence to the actual surroundings by continued experimentation (*DA,* 430a26ff.).

Still, the way in which understanding occurs must be differentiated from cases in which, for example, a rat may solve a maze through repeated iterations. The difference may be the result of the human's distinct capacity to cognize their own understanding (429b29ff.). The mind is increasingly able to recognize and account for the influence of its own perceptive apparatus across multiple iterations of phenomena. It is therefore better able to abstract itself and determine the phenomena themselves, pinpointing recurring phenomenal definitions by way of the implicit or explicit awareness of "the most certain of all principles", that "it is not possible for the same thing at the same time both to belong and not belong to the same thing in the same respect" (*Met.*, 1005b12ff.). The human, then, would exercise a new power of scrutiny, able to speak with certitude regarding the defining features of mazes in general, and not just this particular maze over several attempts. This discovery of the self opens the human animal to new problematics regarding its object of desire. First, it must now consider whether it should be pursued *qua* pleasure, honour, or wisdom (*EN*, 1095bff.). Second, since it is now able to sharpen the horizon separating presently available objects and ones disappearing into a distance of possibility, it must explicitly consider the temporal trajectory it should pursue to achieve its greatest good (*DA*, 443b8-13) These combine to open a nexus of ambivalence in the present, rather than perpetual pursuit.

Knowledge is not only theoretical, but brings with it a productive potency for transforming things into other things (*EN,* 1040a5ff.). In a sense, this is only an extension of the drive of all living things to secure the surrounding things which will preserve the configuration of their mean. Productive knowledge functions by bringing a form in the soul, one such phenomenal determination, into being. Aristotle marks two fundamental moments of enacting a technical capability: first, the human must learn a causal chain by tracing backwards, e.g., "since health is such-and-such, it is necessary, if something is to be healthy, that such-and-such be present, for instance uniformity, and if this is to present there must be warmth, and one goes on thinking continually in this way until one traces the series back to that which, at last, one is oneself [already] capable of making" (*Met.,* 1032b4-10). Second, they must set out on this path until arriving at the desired end by completing the transformation. Although this knowledge must be versatile with regard to the incidental arrangement of materials at hand, it is generalizable and so repeatable once learned. This learning occurs as a sort of "having" for a thing, as if putting on a new coat, adapting its existing sensitivities to phenomena as if newly able to set the object of desire in proximity, within reaching distance (1022b4-8). In sum, then, the living thing wants its desired object; has a determined understanding of the object's phenomenal arrangement, its thinghood; has traced the thinghood back to an existing productive capability; and, then, sets back on the route to doing so when it desires to do so, in the same way that before gaining such an adaptation it would simply reach out and pluck a nearby fruit when wanting. It is this capacity that results in the ersatz field of generation among compound bodies. Once brick-things are produced, they appear as a determinate configuration which can be stacked for the production of a house, which serves to shelter from the elements. The entirety of this field, then, exist as man-made extensions to the human apparatus, which can be developed continuously (cf. *Phys.*, 199a17-19).

The purpose of this extended survey has been to prepare a model of Aristotle's sublunar natural ontology in order to be able to efficiently illustrate the roles of *kinesis* and *dunamis/energeia* *kata kinesis*. The understanding of *kinesis* is hinged on the understanding of *dunamis/energeia kata kinesis*. *Dunamis kata kinesis* is "the origin of a change in another or as another" with regards to the kind of change that is motion. *Energeia kata kinesis* is the enactment, the "being at work", of such a potency. Aristotle expresses *energeia kata kinesis,* in its distinction from *dunamis kata kinesis*, by way of *dunamis meta logon* (productive capability). "What is capable is that which would be in no way incapable if it so happened that the being-at-work of which it is said to have the potency is present" (1047a25-27). Something has the *dunamis meta logon* for walking if, were the corresponding *energeia* present, it would be in no way lacking such a *dunamis*. The definition is circular but, as we are beings in constant possession of such capabilities, Aristotle's meaning is clear to us: I am not walking but if I were to be walking nothing would be impairing me from doing so; I would be ready. I would have the desire, I would have any necessary materials prepared, I would not be being held back by force, and I *would have already secured the versatile understanding required for walking*. In order to be ready to walk, others preconditions notwithstanding, I need to secure the versatile understanding, which involves learning the chain of causes linking the desired object to my existing capabilities, and assimilating this linkage into my cognitive apparatus. Walking involves the following in order of a number of distinct steps in a sensible order to re-institute the integrated causal chain (cf. 1032b15).

Aristotle's meaning is readily intelligible with regards to the phenomenon of motion. *Dunamis kata kinesis* is the ready-to-move – the being-prepared-to-move – and the *energeia kata kinesis* is the moving – the following-through-of-steps – aimed at the assembly of a thing. The *energeia* of a *dunamis* qua *dunamis* is just the setting-to-work of what was prepared, the material; the particular following-through of several determinate steps, aimed for a given rearrangement of such materials. "The being at work of the buildable, *just as buildable*, is building" (*Phys.* 202b9-11). *Rest* is a relatively stable coordination of things (like having ingredients available for a meal); *motion* is the setting them to work, in just such a way (cooking properly), so that the desired thing is transformed out of them (the meal).

We now better able to understand the role of *dunamis/energeia* and *kinesis* in Aristotle's account of nature. The elemental things are locked in perpetual reciprocal transformation from one to another. To the extent that one is, its opposite is in potency (*Gen. et Corr.* 334b910). Here, generation and movement are identical. It is in this perpetual movement/becoming of the elemental things – their process of transforming along their continuums – that distinct things are able to emerge, embodying their mean-states. These bodies variegate the relations between things, opening up potencies for qualitative and local motion.

These realms are, of course, inanimate. There is no control over the enactment of potencies:

> ...some things are capable of causing motion in accordance with reason and their potencies include reason, while other things are unreasoning and their potencies are irrational, the former must being things with souls, but the latter in both kinds of beings; with potencies of the latter sort, it is necessary, whenever a thing that is active and one that is passive in the sense in which they are potential come near each other, that the one act and the other be acted upon... (1048a2-9)

If things are viewed *qua* elemental or compound things, the cause of movement – the enactment of a potency *qua* potency – is akin to the collision of one rock with another inasmuch as the very proximity of something with a corresponding potency is sufficient to trigger a reaction. The inanimate realm operates by necessity – possibility – as much as it does by potency.

With the arrival of living things we see increasing tiers of engagement between the thing and its surrounding phenomena. Plants, themselves compound things, have the potential to transform surrounding things; even while rooted in place, they are loci of motion, maintaining their own elemental integrity by evoking their immediate surroundings into process. They don't merely invert into other things, like the elemental bodies, or bounce off, crush, or serve as the foundation for other things, like the compound bodies, but roughly probe their surroundings and their makeup in order to preserve their own thinghood. While the plant can still be knocked around *qua* its compound body, its movements *qua* its thinghood manifest a new relatedness to surroundings. The plant demonstrates a new freedom of movement in so far as it does not respond only to this or that thing making contact, but to a collection of proximal things.

The sphere of influence belonging to animals extends further. Their several localized calibrations (senses), tuned to given vectors of opposition, allow a more sophisticated survey of surrounding phenomena to serve them in their self-preservation, even allowing for a parallax effect indicating a new dimension of dynamics. The animal's lack manifests in an object of desire appearing within the animal's horizon of sensual awareness. The animal's repeated attempts to capture this object are either consummated or thwarted, gradually refining the animal's sensory apparatus and introducing ambivalence as to the optimal method of pursuit. Animals' ways of pursuit manifest their incorporated sense of the surrounding phenomenal network, and not just immediately, but with the straining toward generalization.

Humans are the first to fully sense the gap between sense-perception and the things perceived, initiating projects of reconciliation. They have a proficiency for picking out and naming patterns in the continuous wash of surrounding phenomena. They bring such patterns into sharp focus, and continuously test and align them against a retained microcosmic model of their surroundings. Their crisp apprehension of their surrounding network allows them to path-forge ways to their objects of desire. First, they must go through a tentative experimental phase of determining connections, and adapt their sensory apparatus accordingly; then they must wait until the appropriate pre-conditions are in place; and then they must follow the path through in precisely such-and-such a way. The latter phase, the enactment of the movement, manifests the human's foregoing globalized understanding of its phenomenal surroundings; the product reflects not only everything that it is, but everything that it is not.

At stake in this description of increasing integration of surroundings is the degree of dimensionality proper to the human world. When humans move, it is not in the same way that a rock moves. The vector of contrariety that characterizes human movement reflects their relatively lucid understanding of their surrounding cosmos. Since human experience is open to various ambivalences, such as the reconciliation of the here and now with indefinite later, the controversy surrounding the good, and the general problem of Being, the products of their doing reflect the variousness of approaches chosen by human-beings in their free movement.

2.

Aristotle engages two challenges to his account: the Megarian and the Protagorean, and the latter only indirectly. The Megarians hold that a *dunamis* has being only when its corresponding *energeia* has being. The potential for a rock to be moving is only there when the rock is actually moving; the potential for a builder to be building is only there when he is actually building. Aristotle does not provide the pathos for this position, but it may be derived.

In Aristotle's account, the onset of vital thinghood and its three tiers, each corresponding to the introduction of a new degree of incorporation of surroundings, and corresponding to the opening of a new dimension of potential interaction, is observed and described, but not explained or "proved". It is *sui generis*. As well, potency is only perceived or deduced at a remove. I see two objects placed near to each other, observe the reactions that tend to occur, and presume the existence of a potency. The phenomena leave room for alternate accounts.

The Megarians offer one such interpretation. There is no real distinction between potency and being-at-work or, in terms of motion, *readiness-to-move* and *following-through-the-steps*. Each are equally things, things with a given composition. Sitting, sitting-up, and standing are each equally ontologically unitary phenomena. By viewing everything in such terms they disregard Aristotle's escalating dimensions of integration with surroundings and corresponding degrees of freedom. All phenomena are treated as compound things. This still permits the study of things and the determination and their network of relations, and so seems non-disruptive of the human project of fully reckoning their surroundings. It also has consequences for the outcome of the project: at the level of interaction between compound bodies, potency coincides with possibility. Any apparent variety of potencies humans seem to enjoy must therefore be epiphenomenal in character, in the way that a confused person might wrongly think a square's diagonal to be potentially commensurable with its side.

Aristotle stakes his defense *qua dunamis meta logon*. All phenomena can not be equally things because capabilities in a definite sense have being and in a definite sense lack it. Insofar as our constantly evidenced distinction between ceasing-to-do and forgetting-entirely must be preserved, *present thinghood* and *being-at-work* must not be collapsed into *being simply*.

Aristotle also remarks that the Megarian position in fact reduces to the Protagorean position. The Protagorean position is more radically opposed to Aristotle's, denying the principle of non-contradiction *(Met.,* 1006a1-4*).* The Protagoreans deny that anything coheres; compound things are not “things” at all. All unifying thinghoods are man-measured products. All there "is" on critical inspection is a furnace of smearing elements (indeed, one that is quick to cover over with the appearance of things). While the Megarian attitude is consistent with the continuation of a (now tantilizingly simplified) project of global map-making, the Protagorean outlook brings a radical problematization of the project of discovery and a new prioritization of the local as the site of original phenomenal encounter. Aristotle maintains a polemic against the Protagoreans, but their position may be held by stressing the phenomenally second-order quality of potencies (cf. Heidegger, 177).

In *Θ* 10, Aristotle identifies the true as “the most governing” way of being. To encounter something in truth is to be in unmistakable contact with it, if to misrepresent it in thought. Truth is that in virtue of which beings are able to emanate in their being to the potential knower. To the extent that such a rendering of truth is taken as securing the ontological priority of thinghood, the issue of the persistent gap between perceiver and perceived is shifted to the margins. Aristotle himself argues against both the Megarians and the Protagoreans. In perhaps prematurely closing the gap between perceiver and perceived, however, he may have unintentionally determined the historical course of the perennial *gigantomachia.*